**1** The Group 17 elements are oxidising agents.

**(a)** **(i)** Explain how the Group 17 elements act as oxidising agents.

.....

..... [1]

**1** Chromium, Cr, and its compounds are widely used in many chemical reactions.

**(a)** Cr exists as four stable isotopes.

**(i)** The most common isotope of Cr is chromium-52.

Determine the number of protons, neutrons and electrons in an atom of chromium-52.

number of protons ..... neutrons ..... electrons .....
[1]

**(c)** Acidified dichromate(VI) ions will convert methanal, $CH_2O$, to carbon dioxide. The movement of electrons to or from relevant species is shown in the following half-equations.

half-equation 1 $CH_2O + H_2O \rightarrow CO_2 + 4H^+ + 4e^-$

half-equation 2 $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

**(i)** Identify the species that is reduced in half-equation 2. Explain your answer.

..... [1]

**(ii)** The oxidation state of the carbon atom in methanal is 0.

Calculate the oxidation state of carbon in carbon dioxide.

..... [1]

**(iii)** Construct the ionic equation for the reaction of dichromate(VI) ions with methanal in acidic conditions.

..... [2]

**1** Manganese, Mn, and its compounds are widely used in many chemical reactions.

**(a)** Mn is usually found as a single isotope, manganese-55.

**(i)** Determine the number of protons, neutrons and electrons in an atom of manganese-55.

number of protons ..... neutrons ..... electrons .....
[1]

**(c)** Manganese(IV) oxide reacts with methanal, $CH_2O$, in acidic conditions to produce carbon dioxide. The movement of electrons to or from relevant species is shown in the following half-equations.

half-equation 1 $CH_2O + H_2O \rightarrow CO_2 + 4H^+ + 4e^-$

half-equation 2 $MnO_2 + 4H^+ + 2e^- \rightarrow Mn^{2+} + 2H_2O$

**(i)** Identify the species that is reduced in half-equation 2. Explain your answer.

.....

..... [1]

**(ii)** The oxidation state of carbon in methanal is 0.

Calculate the oxidation state of carbon in carbon dioxide.

..... [1]

**(iii)** Construct the ionic equation for the reaction of manganese(IV) oxide with methanal in acidic conditions.

..... [2]

**1** Chromium, Cr, and its compounds are widely used in many chemical reactions.

**(a)** Cr exists as four stable isotopes.

**(i)** The most common isotope of Cr is chromium-52.

Determine the number of protons, neutrons and electrons in an atom of chromium-52.

number of protons ..... neutrons ..... electrons .....
[1]

**(c)** Acidified dichromate(VI) ions will convert methanal, $CH_2O$, to carbon dioxide. The movement of electrons to or from relevant species is shown in the following half-equations.

half-equation 1 $CH_2O + H_2O \rightarrow CO_2 + 4H^+ + 4e^-$

half-equation 2 $Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$

**(i)** Identify the species that is reduced in half-equation 2. Explain your answer.

..... [1]

**(ii)** The oxidation state of the carbon atom in methanal is 0.

Calculate the oxidation state of carbon in carbon dioxide.

..... [1]

**(iii)** Construct the ionic equation for the reaction of dichromate(VI) ions with methanal in acidic conditions.

..... [2]

**2** **(a)** Fig. 2.1 shows the covalent bonds and lone pairs of electrons in a molecule of $SCl_2$.

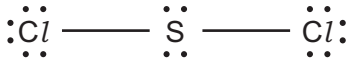


**Fig. 2.1**

**(i)** Use VSEPR theory to predict the shape of a molecule of $SCl_2$.

..... [1]

**(b)** $SCl_2$ is a red liquid that reacts with water.

$$2SCl_2(l) + 3H_2O(l) \rightarrow H_2SO_3(aq) + S(s) + 4HCl(aq)$$

**(i)** Describe **two** observations, other than temperature change, that are made when an excess of water is added to $SCl_2$.

.....

..... [2]

**(ii)** When $SCl_2$ reacts with water, the $SCl_2$ is broken down and a disproportionation reaction occurs.

State the oxidation numbers of sulfur in $SCl_2$ and $H_2SO_3$.

oxidation number of S in $SCl_2$ .....

oxidation number of S in $H_2SO_3$ .....
[2]

**5** The skeletal formula of **W** is shown in Fig. 5.1.

**W**

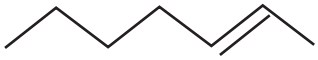


**Fig. 5.1**

**(a)** **W** has two positional isomers. Only one shows stereoisomerism.

Draw the structures of the **two** positional isomers of **W** in the boxes.

[2]

**(d)** **W** reacts with reagent **X** to produce ethanoic acid and pentanoic acid.

**(i)** State the role of **X** in this reaction.

..... [1]

**2** The chemical properties of oxides are related to the chemical bonding present in these compounds.

**(a)** A Period 3 oxide produces a solution with a pH greater than 10 when it is added to water. State the formula of the oxide.

..... [1]

**(c)** Table 2.1 shows the melting points of some oxides.

**Table 2.1**

| oxide | melting point / °C |
|---|---|
| $SO_2$ | –73 |
| $H_2O$ | 0 |
| $SO_3$ | 17 |
| $SiO_2$ | 1610 |
| $Na_2O$ | 1132 |
| MgO | 2852 |
| $Al_2O_3$ | 2072 |

**(i)** Identify the oxide from Table 2.1 that contains the element with the highest oxidation number.

..... [1]

**5** **(a)** **W** is a colourless liquid.

**W**

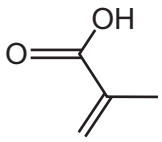



**Fig. 5.1**

**(i)** Deduce the empirical formula of **W**.

..... [1]

**(ii)** **W** reacts with $LiAlH_4$ to produce **Y**.

Draw the structure of **Y**.

[1]

**(iii)** Identify the role of $LiAlH_4$ when it reacts with **W**.

..... [1]

4 **(a)** Propanoic acid, $CH_3CH_2COOH$, reacts with reducing agent **Q** to produce propan-1-ol, $CH_3CH_2CH_2OH$.

**(i)** Suggest the formula of reducing agent **Q**.

..... [1]

**(c)** Compounds **A**, **B** and **C** belong to the alcohol homologous series. Each molecule of **A**, **B** and **C** contains four saturated carbon atoms.

**(i)** Identify the type of hybridisation shown in the saturated carbon atoms of all alcohols.

..... [1]

**(ii)** Identify the gas produced when Na(s) is added to separate samples of each alcohol.

..... [1]

**(iii)** Describe the role of Na(s) when it reacts with alcohols.

..... [1]

**1** **(a)** Solid sodium conducts electricity. Sodium oxide conducts electricity when molten but not when solid.

**(i)** Name the type of bonding present in sodium and in sodium oxide.

bonding in sodium .....

bonding in sodium oxide .....

[1]

**(c)** State the oxidation number of the Period 3 elements bonded to C$l$ in NaC$l$ and PC$l_5$. Explain the difference in the oxidation number.

.....

.....

..... [2]

**(d)** Table 1.2 shows melting points of some oxides.

**Table 1.2**

| oxide | melting point / °C |
|---|---|
| $SO_2$ | –73 |
| $H_2O$ | 0 |
| $SO_3$ | 17 |
| $SiO_2$ | 1610 |
| MgO | 2852 |
| $Al_2O_3$ | 2072 |

A student suggests the following hypothesis.

*The higher the oxidation number of the element combined with oxygen, the higher the melting point of the oxide.*

Use Table 1.2 to deduce if this hypothesis is true or false or if there is **not** enough information to make a conclusion. Explain your answer.

.....

.....

..... [2]

**4** The structure of vitamin C is shown in Fig. 4.1.

vitamin C

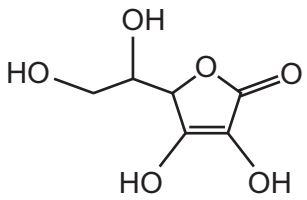



**Fig. 4.1**

**(a)** Deduce the empirical formula of vitamin C.

..... [1]

**(b)** The concentration of vitamin C is found by titration with $I_2$(aq).

vitamin C **Q**

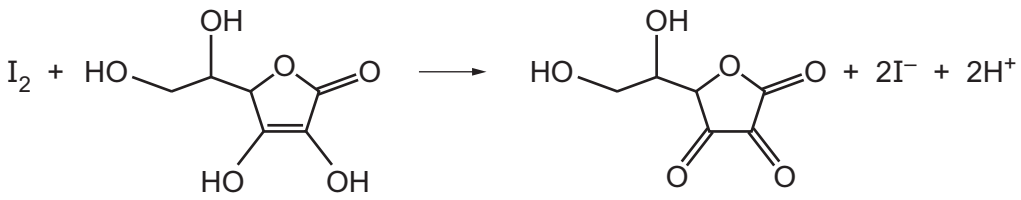



**Fig. 4.2**

A vitamin C tablet is dissolved in water to produce 200.0 $cm^3$ of vitamin C solution.
5.00 $cm^3$ of this vitamin C solution is added to a flask with approximately 150 $cm^3$ of water and an indicator.
Exactly 28.40 $cm^3$ of $5.00 \times 10^{-4}$ mol $dm^{-3}$ $I_2$(aq) reacts with the sample of vitamin C solution in the flask.

[$M_r$: vitamin C, 176]

**(i)** Calculate the amount, in mol, of $I_2$(aq) added to the flask in this titration.

amount of $I_2$(aq) = ..... mol [1]

**(iii)** Deduce the role of $I_2$(aq) in the reaction in Fig. 4.2.

..... [1]

**1** Phosphorus and chlorine are elements in Period 3 of the Periodic Table.

**(a)** Chlorine forms three different compounds with phosphorus.

The most common compounds are $PCl_3$ and $PCl_5$.

**(i)** Complete Table 1.1.

**Table 1.1**

| compound | oxidation number of P | oxidation number of Cl |
|---|---|---|
| $PCl_3$ | | |
| $PCl_5$ | | |

[2]

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**2** Chlorine is one of the elements in Group 17 of the Periodic Table.

**(a) (i)** Describe the colours of the Group 17 elements, chlorine to iodine, at room temperature.

.....

..... [1]

**(iii)** State what is observed when chlorine reacts with hydrogen.

..... [1]

**(ii)** State the essential condition for chlorine to react with $C_2H_6$ at room temperature.

..... [1]

**(iii)** Write **two** equations to show the propagation steps in this reaction.

1 .....

2 .....
[2]

**3** Nitrogen and phosphorus are elements in Group 15 of the Periodic Table.

**(a)** Nitrogen is found in inorganic compounds such as nitrogen oxides ($NO_x$), nitrates and nitric acid.

**(i)** Identify **one** natural and **one** man-made occurrence of nitrogen oxides in the atmosphere.

natural .....

man-made .....
[2]

**(v)** Describe what is seen when solid calcium nitrate is heated strongly.

..... [1]

**(b)** A common test for nitrates is the reaction with NaOH and A*l*. Equation 1 shows the reaction.

equation 1 $3NO_3^- + 8Al + 5OH^- + 18H_2O \rightarrow 3NH_3 + 8[Al(OH)_4]^-$

**(i)** Deduce the oxidation state of nitrogen in $NO_3^-$.

..... [1]

**(ii)** Identify the species that is oxidised in equation 1.

..... [1]

**4** Bromoalkanes are used widely in industry, although there is increasing concern about their environmental impact.

Fig. 4.1 shows a reaction scheme involving 1,2-dibromoethane.

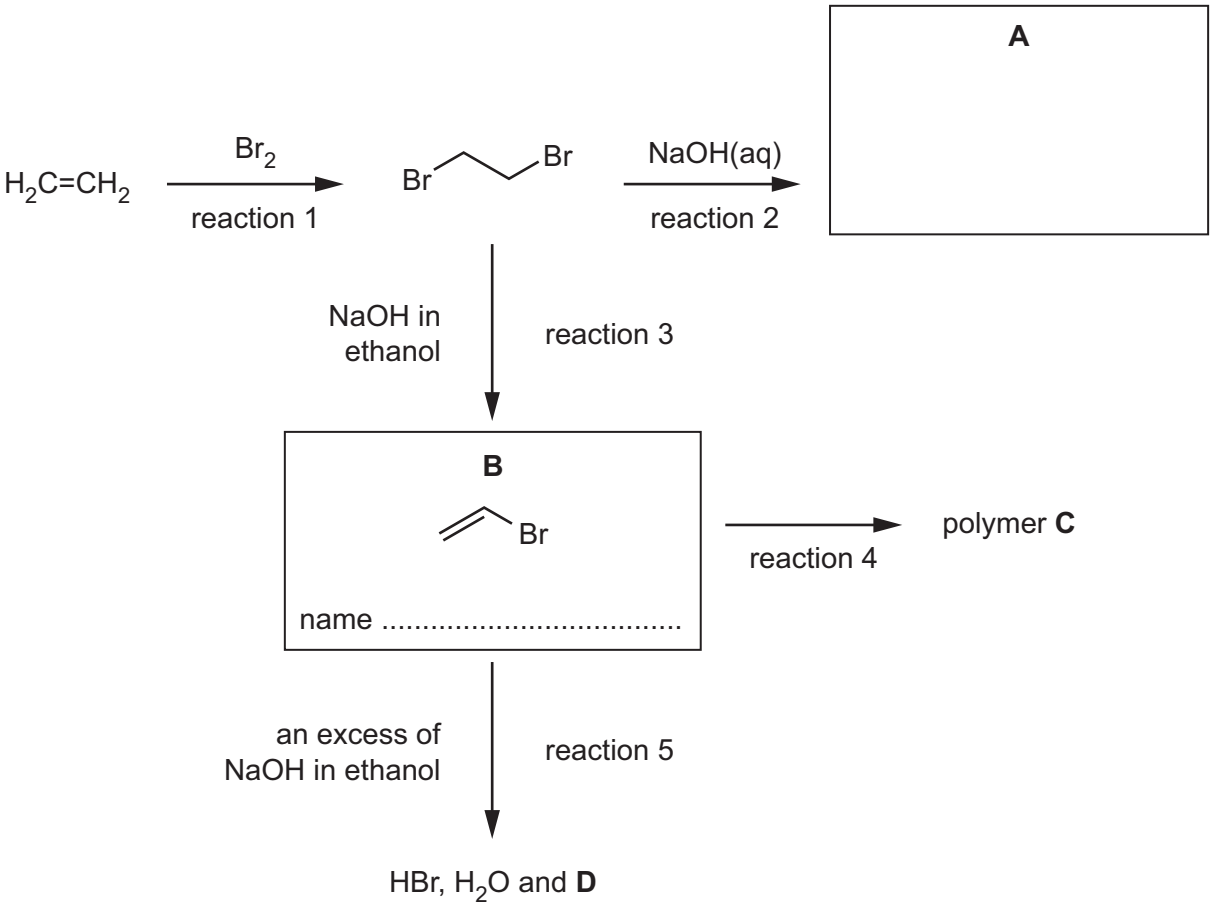



**Fig. 4.1**

**(a)** Complete Fig. 4.2 to show the mechanism for the formation of 1,2-dibromoethane in reaction 1.

Include charges, dipoles, lone pairs of electrons and curly arrows as appropriate.



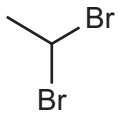

**Fig. 4.2**

[3]

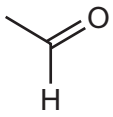
**(d)** Compound **E** is the only isomer of 1,2-dibromoethane.

Alkaline hydrolysis of **E** gives compound **F**.

**E**




**F**




**(i)** Identify the type of isomerism shown by **E** and 1,2-dibromoethane.

..... [1]

**Question 4 continues on page 14.**

**(e)** Compound **F** reacts with reagent **G** to form compound **H**.




The infrared spectrum of **H** is shown in Fig. 4.3.

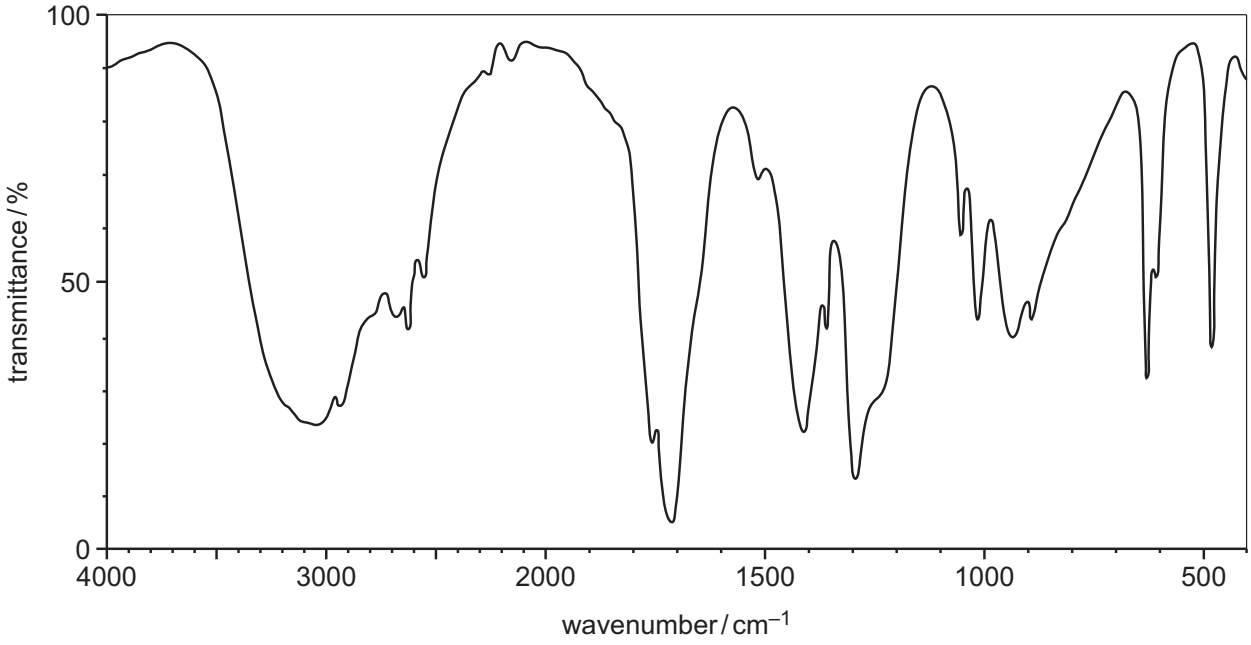



**Fig. 4.3**

**Table 4.2**

| bond | functional groups containing the bond | characteristic infrared absorption range (in wavenumbers) / $cm^{-1}$ |
|---|---|---|
| C–O | hydroxy, ester | 1040–1300 |
| C=C | aromatic compound, alkene | 1500–1680 |
| C=O | amide<br>carbonyl, carboxyl<br>ester | 1640–1690<br>1670–1740<br>1710–1750 |
| C≡N | nitrile | 2200–2250 |
| C–H | alkane | 2850–2950 |
| N–H | amine, amide | 3300–3500 |
| O–H | carboxyl<br>hydroxy | 2500–3000<br>3200–3600 |

**H** also shows a molecular ion peak at *m*/*e* = 60 in its mass spectrum.

**(i)** Use the information in **(e)**, Fig. 4.3 and Table 4.2 to deduce the structure of **H**. Explain your answer fully.

**H**

.....

.....

.....

[3]

**(ii)** Suggest the role of reagent **G**.

.....

[1]

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**2** Chlorine is one of the elements in Group 17 of the Periodic Table.

**(a)** **(i)** Describe the colours of the Group 17 elements, chlorine to iodine, at room temperature.

.....

..... [1]

**(ii)** State the essential condition for chlorine to react with $C_2H_6$ at room temperature.

..... [1]

**(ii)** Define disproportionation.

.....

..... [1]

**1** **(a)** **(i)** Explain the lack of reactivity of nitrogen gas, $N_2(g)$.

.....

.....

..... [2]

**(b)** **(i)** A sample of A$l$ reacts with an excess of C$l_2$.

State the oxidation number of A$l$ in the product of the reaction.

oxidation number of A$l$ ..... [1]

**(d)** The oxide of silicon reacts with calcium oxide in an addition reaction to produce calcium silicate, $CaSiO_3$. The oxidation number of calcium in $CaSiO_3$ is +II.

**(i)** Deduce the oxidation number of silicon in calcium silicate.

oxidation number of silicon ..... [1]

**1** **(a)** The elements of Group 17 are called halogens.

Complete Table 1.1.

**Table 1.1**

| halogen | colour at 293 K |
|---|---|
| chlorine | |
| bromine | |
| iodine | |

[1]

**(c)** Iodine is made by reacting bromine with sodium iodide.

**(i)** Construct an ionic equation for the reaction of bromine with sodium iodide.

..... [1]

**(ii)** State the role of bromine in the reaction. Explain your answer.

.....

..... [1]

**2** **(a)** Sulfur chloride, $SCl_2$, is a liquid at room temperature. When $SCl_2$ is added to water, misty fumes are seen and a solution is made that turns universal indicator red.

**(i)** Identify the type of reaction that occurs when $SCl_2$ is added to water.

..... [1]

**(c)** Solid magnesium nitride, $Mg_3N_2$, is a crystalline solid.

**(i)** Deduce the oxidation numbers of magnesium and nitrogen in magnesium nitride to complete Table 2.1.

**Table 2.1**

| | oxidation number in $Mg_3N_2$ |
|---|---|
| magnesium | |
| nitrogen | |

[1]

**1** Bismuth is an element in Group 15 of the Periodic Table.

**(a)** Bismuth has metallic bonding.

Draw a labelled diagram to show the metallic bonding in bismuth.

[1]

**(c)** $Bi_2O_3$ can be used to form $NaBiO_3$, as shown in equation 1.

equation 1 $\quad Na_2O + Bi_2O_3 + O_2 \rightarrow 2NaBiO_3$

**(i)** Deduce the oxidation number of Bi in $Bi_2O_3$ and in $NaBiO_3$.

oxidation number of Bi:

in $Bi_2O_3$ ..... in $NaBiO_3$ .....
[1]

**(ii)** Identify the reducing agent in equation 1.

..... [1]

**(d)** $NaBiO_3$ is an oxidising agent with similar properties to $KMnO_4$.

Fig. 1.1 shows an example of the use of $NaBiO_3$ as an oxidising agent.

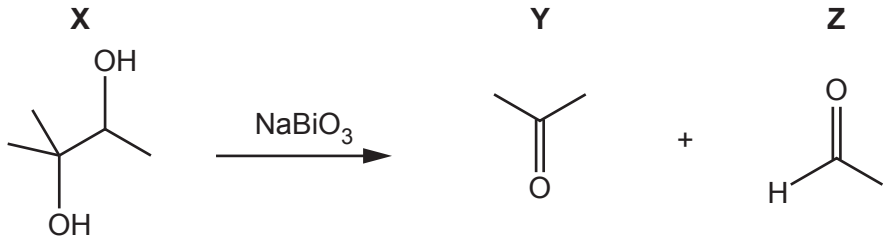



**Fig. 1.1**

**(i)** Explain the term oxidising agent.

.....

..... [1]

**2** Chlorine, $Cl_2$, reacts with many elements and compounds to form chlorides.

Table 2.1 shows information about some chlorides of Period 3 elements.

**Table 2.1**

| | Na | Mg | Si |
|---|---|---|---|
| formula of chloride | | | |
| structure of chloride | giant | | |
| bonding of chloride | | | covalent |
| pH of solution formed on addition of chloride to water | | 6.2 | |

**(a)** Complete Table 2.1. [3]

**(b)** When $Cl_2$ reacts with **cold** NaOH(aq), $Cl_2$ is both oxidised and reduced. The products are NaCl, water and **G**.

**(i)** State the type of redox reaction in which the same species is both oxidised and reduced.

..... [1]

**3** Potassium chlorate, $KClO_3$, is widely used as an oxidising agent and to make $O_2(g)$.

**(a)** Define oxidising agent.

.....

..... [1]

**(c)** When $KClO_3$ is heated without a catalyst, $KClO_4$ and $KCl$ form.

$$4KClO_3 \rightarrow 3KClO_4 + KCl$$

Explain why this reaction is described as a disproportionation reaction.

.....

..... [1]

**3** Phosphoric(V) acid, $H_3PO_4$, is used in both inorganic and organic reactions.

**(a)** $H_3PO_4$ is made in a two-step process from phosphorus.

step 1 Phosphorus reacts with an excess of oxygen to form a white solid.

step 2 The white solid then reacts with water to form $H_3PO_4$.

**(i)** Write an equation for each step.

step 1 .....

step 2 .....

[2]

**(b)** $H_3PO_4$ is also formed in the process shown in reaction 1.

reaction 1 $4H_3PO_3 \longrightarrow 3H_3PO_4 + PH_3$

Table 3.1 shows some relevant thermodynamic data.

**Table 3.1**

| compound | enthalpy change of formation, $\Delta H_f$ / kJ mol$^{-1}$ |
|---|---|
| $H_3PO_3$ | –972 |
| $H_3PO_4$ | –1281 |
| $PH_3$ | +9 |

**(i)** Define enthalpy change of formation.

.....

.....

..... [2]

**(iii)** Explain why reaction 1 is a disproportionation reaction.

Explain your reasoning with reference to relevant oxidation numbers.

.....

.....

..... [2]

**3** Potassium chlorate, $KClO_3$, is widely used as an oxidising agent and to make $O_2(g)$.

**(a)** Define oxidising agent.

.....

..... [1]

**(c)** When $KClO_3$ is heated without a catalyst, $KClO_4$ and $KCl$ form.

$$4KClO_3 \rightarrow 3KClO_4 + KCl$$

Explain why this reaction is described as a disproportionation reaction.

.....

..... [1]

**1** Copper is used in electrical equipment. It has a melting point of 1085 °C.

**(a)** **(i)** Identify the lattice structure of copper.

..... [1]

**(d)** When KI(aq) is added to $CuSO_4$(aq) the blue-coloured solution turns brown and a white precipitate of CuI(s) is seen.

The reaction between copper ions and iodide forms only two products.

**(i)** Complete the equation for this reaction.

.....$Cu^{2+}$ + .....$I^- \rightarrow$ .....CuI + .....

[1]

**(ii)** Identify the oxidising agent in this reaction. Explain your answer in terms of electron transfer.

.....

..... [1]

**2** **(a)** The reaction of pure aluminium is only observed if the aluminium oxide layer is removed first. When pure aluminium is added to cold water, bubbles of gas are seen.

**(i)** State **one** property of aluminium oxide that explains why an aluminium object does **not** react with cold water until the aluminium oxide layer is removed.

..... [1]

**(c)** Aluminium chloride and phosphorus chloride are both white solids.

**(i)** State the maximum oxidation number of aluminium and of phosphorus in these solid chloride salts.

maximum oxidation number of aluminium .....

maximum oxidation number of phosphorus .....

[1]

**2** Chlorine is a reactive element. It forms many compounds.

**(a) (i)** Complete Table 2.1 to show the maximum oxidation number of the elements Na to P in their chlorides.

**Table 2.1**

| element | Na | Mg | A*l* | Si | P |
|---|---|---|---|---|---|
| maximum oxidation number | | | | | |

[1]

**1** Tellurium is an element in Group 16. The most common isotope of tellurium is $^{130}Te$. Its electronic configuration is [Kr] $4d^{10}\ 5s^2\ 5p^4$.

**(a)** Complete Table 1.1.

**Table 1.1**

| | nucleon number | number of neutrons | number of electrons |
|---|---|---|---|
| $^{130}Te$ | | | |

[3]

**(e)** Te reacts with $F_2$ at 150 °C to form $TeF_x$. Molecules of $TeF_x$ are octahedral with bond angles of 90°.

Explain why $TeF_x$ is octahedral with bond angles of 90°.

.....

.....

..... [2]

**(f)** $TeF_x$ reacts with water to form tellurium hydroxide and HF. The oxidation number of tellurium does **not** change during this reaction.

**(i)** Construct an equation for the reaction of $TeF_x$ with water.

..... [1]

**3** Chlorine is a very reactive element.

**(a)** Chlorine reacts with silicon to form silicon(IV) chloride.
Describe the appearance of silicon(IV) chloride at room temperature and pressure. State its structure and bonding.

appearance .....

structure and bonding .....
[2]

**(c)** **(i)** State the reagent and conditions required for the formation of sodium chlorate(V) from $Cl_2$(g).

..... [1]

**(ii)** Explain why the reaction in **(c)(i)** is described as a disproportionation reaction.
Your answer should refer to relevant species and their oxidation numbers.

.....

..... [1]